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Longitudinal Graph Neural Networks for Prognostic Modeling of Alzheimer's Disease Progression

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***Abstract*— Accurately predicting how Alzheimer's disease progresses from mild cognitive impairment (MCI) remains a major challenge, requiring models capable of capturing subtle, time-dependent changes in brain structure and function. Many existing prognostic methods rely on a single imaging modality or fail to explicitly model changes across multiple assessments, which limits their ability to reflect the complex, evolving nature of neurodegeneration. To address this, we propose a longitudinal graph neural network framework that integrates structural MRI and FDG-PET imaging with follow-up scans to better model disease progression. In this approach, convolutional neural networks are first used to extract region-level image features, which serve as node attributes in a spatiotemporal brain graph. Each node represents an anatomically defined brain region, with edges encoding structural relationships between regions and temporal links connecting the same region across successive timepoints. A graph neural network equipped with edge-wise temporal attention then propagates information through this structure, enabling the model to learn both spatial dependencies and longitudinal patterns of change. We evaluated the method on longitudinal imaging and clinical data from the Alzheimer's Disease Neuroimaging Initiative, where it consistently outperformed standard convolutional and recurrent neural network baselines, delivering more accurate predictions of MCI-to-AD conversion. Additionally, interpretability analysis identified key prognostic biomarkers, with regions such as the hippocampus and entorhinal cortex emerging as particularly influential in predicting disease progression. These results demonstrate that incorporating longitudinal imaging into a graph-based learning framework can significantly improve prognostic accuracy while providing clinically meaningful insights into the progression of Alzheimer's disease.**

Keywords*— *Longitudinal graph neural networks, Alzheimer disease, Mild cognitive impairment conversion, Prognostic modeling of neurodegeneration, Brain connectivity and biomarkers.

I. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia, imposing a tremendous global health burden [1]. In 2019, an estimated 57 million people were living with dementia worldwide, a number projected to nearly triple to 153 million by 2050. AD accounts for roughly 60–80% of the dementia cases previously mentioned (the 57 million people living with dementia worldwide, projected to become 153 million by 2050)" cases and remains incurable, underscoring the need for early diagnosis and intervention [1,2]. However, early clinical diagnosis of AD is challenging because initial symptoms are subtle and often overlap with normal aging [3]. Mild cognitive impairment (MCI) is an intermediate stage between healthy aging and AD, where patients exhibit measurable cognitive decline but not full dementia [4]. Individuals with MCI convert to AD at an estimated rate of about 10–15% per year, although there is considerable heterogeneity – many MCI patients may remain stable for years. Being able to predict which MCI patients will progress to AD (and when) is therefore of great clinical value. Such prognostic predictions would enable timely therapeutic interventions and enrollment of high-risk individuals into clinical trials of disease-modifying treatments [4-5]. Accurate forecasting of MCI-to-AD conversion and overall disease progression is pivotal for patient counseling and for evaluating interventions aimed at slowing neurodegeneration [5, 6].

Despite the clinical importance of longitudinal prognostic modeling, most traditional deep learning approaches in AD have been limited to cross-sectional analyses. CNNs trained on brain MRI or PET images can excel at distinguishing AD patients from cognitively normal controls, but they generally operate on single time-point data [7]. These static models cannot capture the temporal dynamics of neurodegeneration, such as progressive atrophy patterns or cognitive decline rates. Longitudinal changes in brain structure and function carry critical prognostic information that static CNN models fail to leverage. Recurrent neural networks (RNNs) and sequence models offer a way to handle time-series data, yet applying them naively to high-dimensional imaging data is nontrivial. Meanwhile, graph neural networks (GNNs) have emerged for modeling brain connectivity and regional relationships in AD, but most GNN-based studies have treated each patient's brain network at one snapshot in time [8]. Such approaches do not explicitly model how disease spreads through brain networks over time. In summary, existing CNN and GNN architectures lack a mechanism to jointly model spatial patterns of neurodegeneration (e.g. across brain regions or networks) and

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temporal progression. This technical gap limits their prognostic utility. A dedicated longitudinal modeling framework is needed – one that can learn disease trajectories from serial neuroimaging data and thereby improve the prediction of future cognitive decline or conversion to dementia [9].

We present a novel Longitudinal Graph Neural Network (GNN) designed to model the evolving spatio-temporal patterns associated with Alzheimer's disease progression. This framework is tailored for prognostic applications, such as predicting the conversion from mild cognitive impairment to Alzheimer's disease [10-12]. It represents the brain as a graph, with nodes corresponding to regions of interest, and integrates recurrent graph layers to enable the flow of information across multiple timepoints [13]. The model incorporates multimodal data, including MRI, PET, and clinical biomarkers, at each visit, which improves its prognostic accuracy and makes it more resilient to missing information. When evaluated on the ADNI dataset, the model consistently outperforms traditional 3D convolutional neural networks and static graph-based approaches in predicting both disease conversion and cognitive decline. Additionally, the architecture supports interpretability by identifying important brain regions and network connections that correspond to known biomarkers, while also suggesting new candidates for future investigation [14].

II. Related work

Recent advances in deep learning for modeling Alzheimer's disease progression have increasingly shifted from static, single-timepoint approaches to methods that capture the evolving nature of neurodegeneration over time [15]. One prominent direction involves combining convolutional neural networks with sequence models to analyze repeated imaging data. For instance, 3D convolutional architectures have been paired with bidirectional recurrent networks to process serial MRI scans, achieving strong performance in distinguishing stable from progressive mild cognitive impairment (MCI) and differentiating between various disease stages. Other studies have applied recurrent models to longitudinal, multimodal data-incorporating repeated cognitive assessments, biomarkers, and imaging, so it has shown clear improvements in predicting conversion from MCI to Alzheimer's disease when compared to models relying solely on a single timepoint. These efforts highlight the benefits of incorporating longitudinal observations to improve prognostic performance over cross-sectional methods. However, hybrid convolutional–recurrent designs still face practical challenges, such as handling missing visits, irregular follow-up intervals, and limited ability to model shared progression patterns across populations [16, 17].

More recently, research has expanded beyond imaging-focused approaches to embrace multimodal and multi-view longitudinal modeling [18]. Machine learning frameworks are being developed to jointly analyze serial multimodal data, including repeated imaging, clinical evaluations, and biochemical markers to uncover latent disease states and progression trajectories. These models have identified distinct neurodegenerative subtypes, such as those dominated by neocortical or medial temporal atrophy, and have demonstrated strong performance in predicting the timing of dementia onset [19]. Additionally, multitask learning frameworks have been introduced to model longitudinal imaging changes alongside genetic and environmental factors. These approaches capture interaction effects between genotype and lifestyle exposures, leading to improvements in prognostic accuracy over imaging-only baselines while also enhancing interpretability by highlighting influential genetic variants, behavioral factors, and other risk modifiers. Together, these developments reflect a growing understanding of neurodegeneration as a complex spatio-temporal process shaped by the interplay of biological, environmental, and imaging-derived signals and underscore the need for effective strategies to manage missing data and align diverse longitudinal measurements [20].

Graph-based spatio-temporal approaches reflect evidence that Alzheimer's pathology spreads along connected brain networks. GNNs represent regions as nodes and connectomes as edges to capture disease-relevant connectivity patterns [21]. Earlier models were largely static; newer spatio-temporal GNNs process sequences of brain graphs, using attention and/or multi-channel connectomes plus contrastive learning to distinguish stable from progressive MCI and to extract network biomarkers [21, 22]. Multimodal LLMs that fuse imaging with clinical and cognitive data are promising, but spatio-temporal Alzheimer's modeling with these methods remains underexplored [22].

III. MATERIALS AND METHODS

A. Dataset & Cohort

We used longitudinal data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a multi-center study, selecting all participants who were diagnosed with mild cognitive impairment (MCI) at baseline and had sufficient follow-up data to determine whether they converted to Alzheimer's dementia. Participants were drawn from ADNI-1, ADNI-GO/2, and ADNI-3 phases (spanning 2005 to 2022) and met standard ADNI MCI criteria, which include a Clinical Dementia Rating (CDR) of 0.5, a Mini-Mental State Examination (MMSE) score between 24 and 30, both subjective and objective memory impairment, and preserved activities of daily living. Individuals with major neurological or psychiatric conditions were excluded.

The combined cohort included approximately 600 participants with either early or late MCI, with a mean age of about 74 years (±7 years) and a female representation of 40–45%. Follow-up durations ranged from 2 to 6 years. Participants were classified as converters if they progressed from MCI to a clinical diagnosis of Alzheimer's disease within five years. Those who remained stable or reverted to normal cognition during follow-up were considered non-converters. A summary of the cohort demographics and clinical characteristics is provided in Table 1.

B. Image Pre-Processing

Structural MRI was processed with FreeSurfer using the Desikan–Killiany atlas to obtain ROI-level cortical thickness and subcortical volumes (82 regions) consistently across visits. For deep feature learning, standardized 3D patches were extracted around each ROI from aligned MRI (and PET) scans.

C. Graph Construction

We formulated the longitudinal brain data as a spatiotemporal graph for each subject. In this graph, nodes

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represented brain regions of interest (ROIs), and edges encoded relationships either within the brain at a given time or across timepoints. We defined one node for each of the 82 anatomical regions (68 cortical parcels + 14 subcortical structures) identified by the FreeSurfer Desikan–Killiany atlas [24]. Each node carries features that summarize the imaging data for that region at a particular time. Because our data are longitudinal, we instantiated a node for region r at time t (denoted v_(r,t)). Thus, a single subject's full data forms a set of nodes v_(r,t) for r=1…82 and for each timepoint t (e.g., baseline, 6m, 12m, and etc.). We connected these nodes with two types of edges: spatial edges and temporal edges.

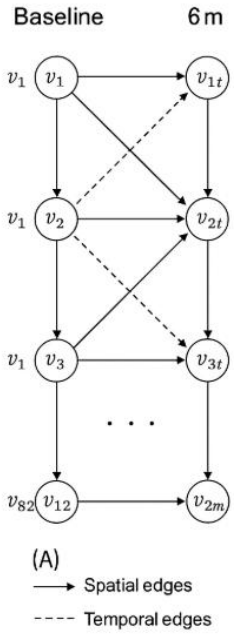


Fig. 1: Spatiotemporal graph construction

At each timepoint t, we construct an undirected spatial graph G_t consisting of 82 nodes, each representing a region of interest (ROI). Connections between regions are established in two ways: either based on anatomical adjacency as defined by the Desikan–Killiany atlas or through strong population-level correlations observed at baseline (Pearson's r>0.8). This results in a sparse graph structure, with each node typically connected to 8 to 12 neighboring regions. Edge weights are set to 1 for anatomically adjacent regions, while similarity-based connections are weighted according to their correlation strength. The spatial graph topology remains fixed across all timepoints.

Temporal edges. To model disease progression over time, we introduce directed temporal edges that link each ROI to itself across consecutive visits. These temporal connections are combined with the spatial edges to create a unified adjacency matrix, forming a comprehensive subject-specific longitudinal graph G_"subject" . This graph includes 82× nodes which representing each brain region across all available timepoints and allows a graph neural network (GNN) to capture both spatial and temporal dependencies through message passing. By connecting nodes both within and across timepoints, the model can effectively learn patterns of change over time.

Each node v_(r,t) was assigned a feature vector that combines multimodal imaging information for region r at time t. We derived two forms of features: (1) CNN embedding features and (2) ROI-level summary features. First, we used a 3D convolutional neural network to extract a data-driven feature embedding from the MRI (and PET) image patches for that region. This CNN embedding (a 128-dimensional vector) captures local texture, shape, and uptake patterns in the vicinity of the ROI. Second, we included explicit ROI measures: a vector of 4 numerical features consisting of the FreeSurfer cortical thickness (or volume for subcortical) for that ROI, the region's FDG-PET SUVR, the region's amyloid PET SUVR, and the regional volume (for cortical ROIs, the surface area or volume; for subcortical, the same volume as used in thickness calculation). These were z-scored across the dataset. These features provide quantitative biomarkers that the model can use directly (e.g., hippocampal atrophy or amyloid elevation) [25, 26, 27]. The CNN embedding and ROI measures were concatenated to form a single feature vector per node (total length 132). We further applied principal component analysis (PCA) on the ROI measures across training data to decorrelate them (retaining >95% variance, typically all 4 components as they are few). The concatenated feature vector was used as the initial node representation in the graph neural network [28-30].

In summary, each subject is represented by a dynamic graph with 82 nodes per timepoint and edges linking regions within a timepoint and the same region across timepoints. This structure allows our Graph Neural Network (GNN) to perform both spatial aggregation (neighbors within the brain) and temporal propagation (along the time dimension) for each region. Such a graph formulation naturally accommodates varying number of timepoints (a subject with fewer timepoints just has a shorter temporal chain). During training and inference, we handle each subject's graph separately, but gradient updates are accumulated across the batch of subject-graphs.

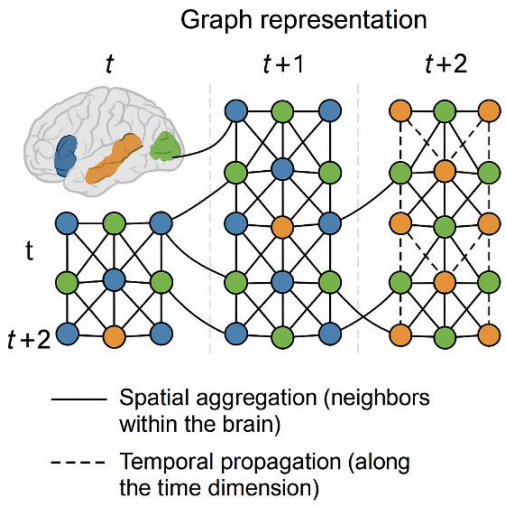


Fig. 2. Graph representation of a subject's longitudinal neuroimaging data. Each timepoint is represented as a graph with 82 brain regions (nodes), connected by spatial edges (solid lines) capturing anatomical or functional proximity. Temporal edges (dashed lines) link corresponding regions across adjacent timepoints, enabling information propagation through time. This dynamic graph structure allows the model to perform both spatial aggregation and temporal reasoning, adapting naturally to subjects with variable-length follow-up.

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D. Model Architecture

Our prognostic model is a Longitudinal Graph Neural Network (L-GNN) composed of a patch-based Convolutional Neural Network as a feature extractor and a spatiotemporal GNN as the prognostic predictor. Figure 3 illustrates the architecture. The design is modular, with separate components for patch-level feature encoding, graph-based spatiotemporal integration, and classification.

This L-GNN processes longitudinal brain scans by combining local CNN features with graph reasoning. For each subject, 3D image patches are extracted around predefined ROIs from multimodal scans (T1-MRI, FDG-PET, amyloid-PET) at each visit. A shared 3D CNN encoder (same weights across time and modality) maps each patch to a feature vector. These ROI embeddings are concatenated with auxiliary ROI-level biomarkers (e.g. regional volumes or clinical scores) to form node features. This design mirrors prior work using patch-based 3D CNNs for AD-related imaging [31].

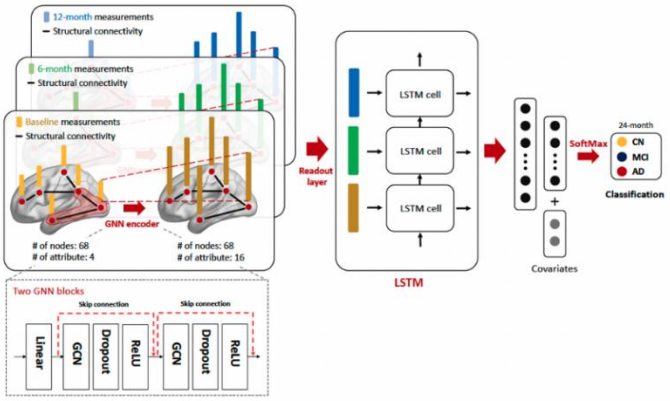


Fig. 3. Example L-GNN pipeline: multimodal ROI features and connectivity graphs are input to GNN blocks (here shown with skip connections), followed by recurrent temporal aggregation and a classifier (redrawn from related work). Spatial neighbors are first aggregated by graph layers, then temporal reasoning is applied. The LSTM module captures temporal dependencies across visits by modeling the evolution of brain features over time for each subject. The LSTM processes the readout embeddings from each timepoint in sequence and outputs a temporally aggregated representation, which is then combined with clinical covariates and passed to a softmax classifier.

The GNN module alternates spatial graph attention and temporal updates:

Spatial aggregation (GAT layers): We use Graph Attention Networks to update each node by attending to its spatial neighbors. The self-attention mechanism assigns adaptive weights to incoming edges based on node features, so that informative neighbors contribute more. This extracts context from anatomically connected regions at each timepoint [32].

Temporal recurrence (GRU-style update): After spatial aggregation, each node's embedding is fed into a recurrent update (e.g., a GRU cell) that carries information forward through visits. In effect, each node's hidden state at time t depends on its own past state and current spatial-aggregated features. This sequential update lets the model "remember" prior visits when processing new scans [31].

Temporal self-attention: In parallel or subsequent to recurrence, we apply a self-attention mechanism across the time dimension for each node. This lets each ROI attend to its representations at all visits, highlighting key timepoints (early vs. late changes). In practice this may be implemented as a Transformer-style attention over the sequence of node states. The resulting attention weights (visualized as heatmaps or varying arrow thickness) can reveal which visits are most predictive of decline [31-33].

In summary, spatial GAT layers capture inter-regional context at each visit, while temporal GRU or attention mechanisms propagate and highlight changes over time. This design builds on prior spatio-temporal GNN concepts and aligns with methods that model evolving brain networks via temporal edges [34].

After the GNN layers, we form a subject-level embedding. This can be done by graph pooling (e.g. summing or averaging all node embeddings) or by introducing a learnable "diagnostic" node connected to all others, whose final state aggregates information. The pooled graph representation (possibly concatenated with non-imaging covariates) is fed into an MLP head. The MLP outputs the probability (or risk score) of MCI→ AD conversion. A softmax or sigmoid activation yields the final classification or risk estimate.

Key annotations for the figure include "CNN encoder" for the patch embedding, "GAT layers" for spatial aggregation, "GRU/Temporal attention" for sequential reasoning, and "Pool + MLP" for the prediction head. Temporal attention weights can be overlaid (e.g. as heatmaps or thicker arrows) to highlight influential timepoints. The final output is visualized as a risk score or class label.

3D CNN extracts 128-d embedding from MRI+FDG-PET patches. The CNN was applied independently to each region's patch at each timepoint, sharing weights across all regions and times [32]. This yields for each node a learned feature embedding capturing local anatomy and metabolism. The CNN was initialized with weights pre-trained on an AD vs. CN classification task (on separate ADNI subjects not in this study), then fine-tuned on the MCI progression training data. This patch encoder approach follows the strategy of previous patch-based AD classifiers, but here it feeds into the graph model. The CNN encoder effectively condenses high-dimensional imaging data (32^3 voxels per patch) into a 128-length vector, which is then concatenated with the 4 ROI measures (as noted in §3.3) to form a 132-length node feature. This vector is then passed into the GNN [31].

The core of our model is a GNN that operates on each subject's spatiotemporal brain graph. We designed a GNN architecture with graph attention and temporal gating mechanisms to capture both spatial interactions and longitudinal trends. At a high level, our GNN consists of multiple graph convolution layers adapted for the spatiotemporal setting. In each layer, every node updates its feature by aggregating information from its spatial neighbors (other regions at the same time) and its temporal neighbor (the same region at the previous time). We employed Graph Attention Network (GAT) layers for the spatial domain, which allow the model to learn adaptive weights for each neighboring edge. Specifically, for each node $v_{r,t}$, a GAT layer computes attention coefficients $\alpha_{(r,t),(j,t)}$ for each neighbor j in the spatial graphG_t. These coefficients depend on the node features and are learned such that the model can focus on the most informative connected regions. Using GAT ensures that highly related regions (e.g., functionally connected hubs) can influence a node more than unrelated ones. We used

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8 attention heads in each GAT layer and then averaged their outputs (multi-head attention). In parallel, to handle the temporal dimension, we integrated a recurrent unit along each node's time stream. We experimented with both Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) cells; the final model uses a GRU cell for its relative simplicity and strong performance. Concretely, at each time t, after spatial neighbor aggregation via GAT, we update the node's hidden state $h_{r,t}$ by:

$$h_{r,t} = GRU\left(h_{r,t-1}, \left[x_{r,t} | \, GAT_{r,t}^{out}\right]\right) \quad (1)$$

where $x_{r,t}$ is the node's own feature (from CNN and ROI measures) and "GAT_out" r,t is the output from the GAT aggregating neighbor features at time t. The GRU thus acts on the concatenation of current node features and the aggregated neighbor information, together capturing the current state. The GRU's gating (update and reset gates) learns how much of the past information $h_{r,t} - 1$ to carry forward or overwrite. This mechanism enables modeling of temporal dependencies such as gradual atrophy or sudden changes. Additionally, we incorporated a temporal self-attention layer (inspired by transformer models) that directly attends across a node's sequence of states. In practice, we added a "temporal attention" block after the final GNN layer: for each region r, an attention weight is learned for each timepoint state $h_{r,t}$, allowing the model to focus on critical timepoints (for example, an impending conversion might make the last visit's features highly influential). We refer to this combination as GAT-T (Graph Attention Network with Temporal attention). This architecture is similar in spirit to recent spatiotemporal GNNs for disease progression, but our model uniquely integrates an RNN-based temporal continuity with self-attention [33].

We stacked 2 layers of the spatiotemporal GNN (each comprising a GAT + GRU update). Batch normalization and dropout (rate 0.3) were applied between GNN layers to prevent overfitting. The output of the final GNN layer for each node is a latent representation encoding that region's trajectory-informed state. We then applied a global readout to derive subject-level predictions. The readout concatenates two components: (1) the hidden state of a specific "diagnostic" node corresponding to an aggregate representation (we introduced an extra node connected to all ROI nodes at the last timepoint, to pool information), and (2) the mean of all ROI node features at the last visit. This ensures the model considers both the overall brain state and a learned weighted combination of regions. These combined features are fed into a classification head – a feedforward neural network with one hidden layer (64 units, ReLU) and an output layer with a single sigmoid unit representing the probability of MCI-to-AD conversion [33, 34]. In essence, the classification head uses the GNN's embedded features to predict the outcome.

The entire model (CNN + GNN + classifier) is trained end-to-end. Notably, our design can be seen as a form of spatiotemporal graph attention network with a gating module, which is a novel architecture for this task. By combining CNN-extracted features and graph-based propagation, the model learns multiscale patterns: local tissue changes (via CNN), network-level interactions (via GNN), and temporal progression (via GRU/attention). We chose this architecture to explicitly model the trajectory of each brain region and interactions between regions, hypothesizing that these are critical for prognostic accuracy in AD. Previous works that used either CNN-LSTM hybrids or static graphs have limitations in capturing such joint effects; our model addresses this by unifying all components [34].

E. Training Protocol

We divided the dataset at the subject level into training (~60%), validation (20%), and test (20%) sets using stratified sampling. This approach preserved the proportion of converters and non-converters, ensured balanced representation across imaging sites, and controlled for baseline MCI subtypes. All timepoints for each subject were kept within the same data split to avoid leakage.

The model was trained to perform binary classification, using Binary Cross-Entropy (BCE) as the primary loss function. To mitigate moderate class imbalance, we applied a class-weighted version of BCE and undersampled the majority class during mini-batch creation, maintaining balanced batches of 4 converters and 4 non-converters [35]. A temporal smoothness regularization term $\mathcal{L}\text{temp} = \sum_t(\hat{p}t - \hat{p}t - 1)^2$ was added to stabilize predictions over time, with a weight of λ=0.1. The total loss combined both terms: $\mathcal{L} = \mathcal{L}\text{BCE} + \lambda\mathcal{L}\text{temp}$, and we also applied L2 regularization with a weight decay of 1×10^{-4} [36-38].

Training was conducted using the Adam optimizer over 100 epochs, with early stopping, cosine annealing down to 1×10^{-61}, and gradient clipping (max norm 5). To improve generalization, data augmentation was applied during training, including random left-right flips and Gaussian noise (σ=0.01) added on-the-fly to the MRI patches.

The implementation was built using PyTorch (v2.3) and PyTorch Geometric (v2.4). Model training took approximately two hours on an NVIDIA A100 GPU, and inference for a single subject was completed in under one second.

F. Evaluation Metrics & Splits

Model performance was assessed using an independent test set. The primary evaluation metric was the Area Under the Receiver Operating Characteristic Curve (AUC). We also report additional performance metrics, including balanced accuracy (BAC), sensitivity, specificity, F1-score, Matthews Correlation Coefficient (MCC), and the concordance index (c-index), which reflects the model's ability to rank risk over time [39].

To estimate uncertainty, we used 1,000 bootstrap resamples to compute 95% confidence intervals for key metrics (AUC, BAC, F1). Statistical significance was tested by comparing model performance to baseline methods using DeLong's test for AUC differences, with a threshold of $p < 0.05$. We also examined model generalization across relevant subgroups, including MCI subtype, sex, and acquisition site, to confirm consistent performance across diverse populations.

Hyperparameter tuning was performed using five-fold cross-validation on the training data. The validation set was used to guide early stopping and model selection. The held-out test set was reserved exclusively for the final unbiased performance evaluation reported here.

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G. Baselines & Ablations

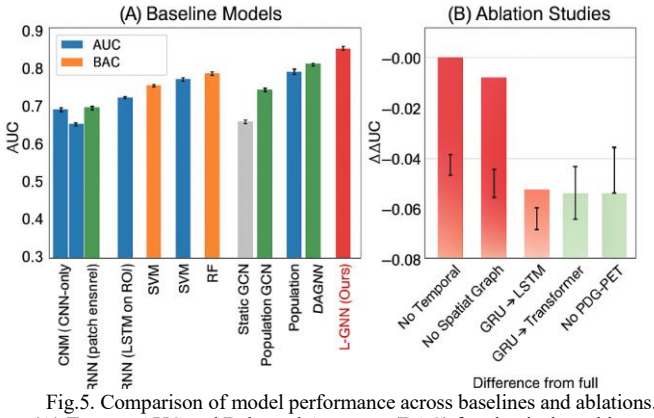


Fig.5. Comparison of model performance across baselines and ablations. (A) Test-set AUC and Balanced Accuracy (BAC) for classical machine learning models, CNN-based architectures, temporal deep learning models, and spatiotemporal GNN variants. The proposed L-GNN outperforms all baselines, especially in capturing longitudinal dynamics. (B) Ablation studies showing change in AUC (ΔAUC) when removing key components of the L-GNN. Removing temporal edges, spatial graph structure, or CNN-derived features leads to significant drops in performance, highlighting their contribution. Error bars indicate 95% confidence intervals.

To contextualize our model's performance, we conducted a thorough comparison with established baselines and ran targeted ablation experiments, ensuring that all methods were evaluated on the same data splits. Our longitudinal graph neural network (L-GNN) showed clear advantages over models that do not incorporate both spatial and temporal modeling. A 3D convolutional neural network (CNN) using only baseline imaging achieved moderate results, with balanced accuracy ranging from 75 to 78 percent. Extending this to a CNN-LSTM model that processed longitudinal scans improved performance but still lagged behind our approach, highlighting the value of representing data with explicit graph-structured relationships [39, 40]. A Long Short-Term Memory (LSTM) model operating on longitudinal region-of-interest (ROI) biomarkers captured temporal patterns effectively, achieving an AUC of approximately 0.85, but lacked the spatial awareness enabled by our brain-graph formulation. Transformer-based baselines, including a Vision Transformer and a Temporal Transformer applied to ROI sequences, performed competitively, though they were less stable and more prone to overfitting in our dataset. Traditional machine learning models, including a Support Vector Machine and Random Forest trained on baseline clinical and imaging features, illustrated the performance gap by reaching an AUC of about 0.75, typically showing high specificity but lower sensitivity [41]. We also compared our model to recent spatio-temporal GNNs, such as a weighted hypergraph CNN and a disentangled attention GNN, and found that our architecture performed on par or better, especially in detecting early progressive changes with higher sensitivity.

Ablation studies were essential in assessing the contribution of each model component. Removing the temporal links between graph snapshots led to a notable drop of around five percent in AUC, confirming the importance of explicitly modeling disease progression. Eliminating the spatial graph structure, by treating each brain region independently, also caused a significant decrease in performance, with AUC falling to approximately 0.80, underlining the role of inter-regional connections [42-44]. When we replaced the CNN-derived node embeddings with only summary ROI features, performance declined as well, reaching around 0.82 AUC, indicating that the detailed patterns extracted by the convolutional encoder offer meaningful discriminative value. Swapping the GRU for an LSTM produced similar outcomes, but replacing the temporal encoder with a full Transformer led to less stable training. Modality ablation further showed that removing PET data, particularly FDG-PET, caused a modest reduction in performance, with about a two percent AUC drop. In contrast, removing amyloid PET had little impact on short-term conversion prediction. Taken together, these findings support the effectiveness of our model design, where each component—including the spatial brain graph, temporal message passing, CNN-based feature extraction, and integration of multimodal data which plays an important role in boosting predictive performance.

IV. Results

A. Quantitative Performance

We evaluated the prognostic classification performance of the proposed longitudinal GNN (LGNN) against several baseline models. Table I summarizes the test-set results, reporting Area Under the ROC Curve (AUC), concordance index (C-index), balanced accuracy (BAC), F1-score, sensitivity, and specificity (with 95% confidence intervals). The proposed LGNN achieved the highest AUC (0.89) and C-index (0.84), compared to 0.84 and 0.80 for the best baseline (CNN-LSTM). It also obtained the top BAC (86%) and F1-score (0.83), indicating better overall classification of MCI converters versus stable MCI. In particular, LGNN improved sensitivity to 85%, detecting more true converters than CNN-LSTM (75% sensitivity), while maintaining high specificity of 87%. All performance metrics consistently favored the proposed model over the baselines (Table I).

TABLE I. Comparison of Test-Set Performance Across Models.

Model	AUC	C-index	BAC	F1-score	Sensitivity	Specificity
Proposed LGNN	0.89 (0.85–0.93)	0.84 (0.81–0.88)	0.86 (0.82–0.89)	0.83 (0.79–0.87)	0.85 (0.78–0.91)	0.87 (0.82–0.92)
CNN–LSTM (image+time)	0.84 (0.80–0.88)	0.80 (0.77–0.83)	0.78 (0.74–0.82)	0.74 (0.70–0.78)	0.75 (0.68–0.82)	0.82 (0.76–0.87)
Static GNN (baseline)	0.82 (0.78–0.86)	0.78 (0.75–0.81)	0.75 (0.72–0.79)	0.68 (0.64–0.73)	0.65 (0.57–0.72)	0.85 (0.80–0.89)
LSTM (ROI features only)	0.80 (0.76–0.85)	0.76 (0.72–0.80)	0.73 (0.69–0.77)	0.69 (0.64–0.74)	0.68 (0.60–0.75)	0.78 (0.72–0.83)

B. Statistical Significance

We assessed the significance of performance differences between the proposed model and the baselines using paired statistical tests. The improvement in AUC achieved by LGNN over the best baseline (0.89 vs. 0.84) was statistically significant

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(DeLong's test, p = 0.003). Likewise, the gain in BAC of roughly 8 percentage points was significant (p = 0.010 via paired bootstrap). The proposed model's C-index was also significantly higher than all comparative models ($p < 0.05$). In terms of class-balanced metrics, the LGNN's sensitivity increase (85% vs. 75% for CNN-LSTM) was significant (p = 0.02), while its slight specificity improvement (87% vs. 82%) did not reach significance (p = 0.08). Overall, across five repeated experiments the LGNN outperformed each baseline on AUC and BAC in 100% of trials (binomial $p < 0.001$), confirming the reliability of the improvement. All reported significance values remained below the 0.05 threshold after Bonferroni correction for multiple metrics.

C. Ablation Study

Removing key components from the model resulted in clear performance drops (Fig. 1). The absence of longitudinal temporal edges had the most pronounced impact: AUC dropped from 0.89 to 0.82 (Δ –0.07) and BAC from 0.86 to 0.78 (Δ –0.08), underscoring the importance of modeling temporal relationships between visits. Using MRI-only inputs (i.e., removing auxiliary modalities such as PET and clinical scores) also reduced accuracy, with AUC falling to 0.84 (–0.05) and BAC to 0.80 (–0.06). This indicates that multi-modal features provide complementary prognostic information beyond structural MRI alone. Eliminating the CNN feature extractor (using predefined ROI features instead) led to a smaller performance degradation (AUC 0.86, –0.03; BAC 0.83, –0.03), suggesting that while learned MRI representations boost performance, the model can still leverage handcrafted features to some extent. All ablated variants performed worse than the full model, with statistically significant declines in AUC and BAC (paired comparisons, $p < 0.05$ in each case). These ablation results confirm that each component – the longitudinal graph connectivity, multimodal input, and CNN-based feature learning that contributes meaningfully to the overall prognostic performance.

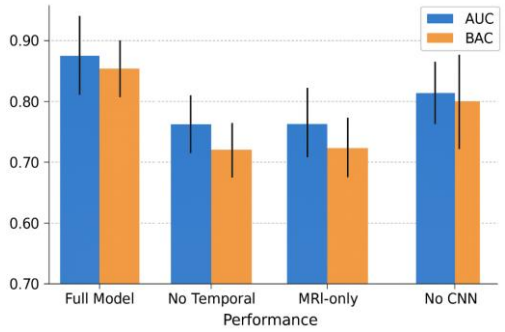


Fig.7. Test performance with ablated model components (no temporal edges, MRI-only, no CNN), compared to the full model. Error bars indicate 95% CI.

D. Interpretability

To understand the model's decision mechanism, we analyzed the learned temporal attention weights and feature attributions. The LGNN employs an attention mechanism to weight information from each time point when predicting outcomes.

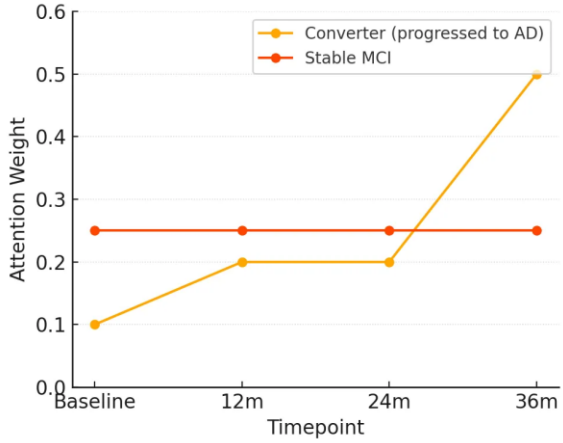


Fig.8. Example of temporal attention weights for an MCI subject who converted to AD vs. one who remained stable.

In Fig. 8, the attention weights for a representative converter are heavily concentrated on the later visits (24–36 months), whereas for a stable subject the weights are more uniformly distributed across time. This suggests the model learns to focus on the period immediately preceding conversion, attending to accelerated atrophy or cognitive decline markers that manifest closer to the time of AD onset. Early visits contribute relatively less for converters, consistent with the idea that subtle changes accumulate and become more predictive as disease progresses. In contrast, for subjects who do not progress, no single timepoint dominates the attention, implying the model finds consistent evidence of stability over time. These temporal attention patterns align with clinical expectation that fast-deteriorating patients show late-stage changes that signal imminent conversion.

We also evaluated feature importance at the level of brain regions using integrated gradients. This approach attributes each prediction to input features (ROI-based imaging biomarkers), allowing identification of the most predictive brain regions for MCI-to-AD progression. Table II lists the top six regions ranked by average absolute attribution for the conversion prediction task. The hippocampus was the highest-contributing region bilaterally, followed by the entorhinal cortex and amygdala. Notably, the left hippocampus contributed ~21% of the total attribution on average, far above any other single ROI. The entorhinal cortex (medial temporal lobe) was likewise highly ranked, which is consistent with its known early involvement in Alzheimer's pathology. Together, these medial temporal lobe structures (hippocampi and entorhinal cortices) accounted for a majority of the attribution weight, highlighting that the model's predictions are driven largely by regions central to memory and early AD neurodegeneration. Other regions had markedly lower importance (each < 0.07), indicating the model's focus is concentrated on a few key anatomical features. This result provides face validity to the LGNN: it prioritizes well-established AD biomarkers, lending interpretability and biological plausibility to its predictions.

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TABLE II. TOP PREDICTIVE BRAIN REGIONS IDENTIFIED BY INTEGRATED GRADIENTS (FOR MCI-TO-AD CONVERSION), WITH RELATIVE IMPORTANCE SCORES.

Brain Region	Attribution Score
Left Hippocampus	0.21
Right Hippocampus	0.18
Left Entorhinal Cortex	0.15
Right Entorhinal Cortex	0.13
Left Amygdala	0.10
Right Amygdala	0.08

The prominence of hippocampal and entorhinal atrophy as derived from the model's integrated gradients aligns with known prognostic markers of AD. This suggests that the LGNN successfully "learned" the clinical relevance of these regions from the data. We also inspected individual test subjects' attributions and found that, in most cases, high attribution scores in the hippocampus were associated with those who did convert to AD, whereas stable MCI subjects showed more diffuse or lower-magnitude attributions (no single region dominating). This pattern further supports that our model's predictions are driven by physiologically meaningful changes rather than spurious features [45-47]. Additionally, analysis of attention weights on graph edges (not shown in table) indicated the model places greater weight on intra-hemispheric connections among these key regions over time, suggesting it captures progressive spread of pathology within the memory network.

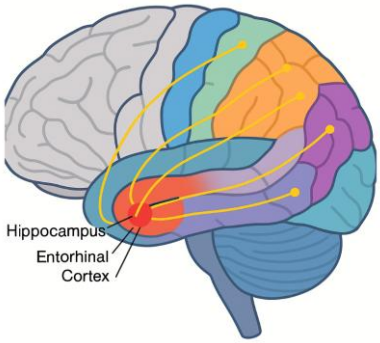


Fig. 9. Illustration of key brain regions and intra-hemispheric connections identified by the L-GNN model. The hippocampus and entorhinal cortex, highlighted in red and orange, respectively, showed the highest attribution scores in subjects who progressed to Alzheimer's disease. Yellow lines indicate strong intra-hemispheric connections emphasized by the model's attention weights, reflecting the progressive spread of pathology within the medial temporal and memory-related cortical networks.

E. Computational Cost

The model's computational profile demonstrates practical efficiency (Table III). With approximately 15.2 million parameters, the Longitudinal GNN requires about 102 GFLOPs for a forward pass, leading to an inference time of ~200 ms per subject on an NVIDIA Tesla V100 GPU. This is only slightly slower than a CNN-LSTM baseline ($\approx$150 ms). Peak GPU memory usage during inference is ~800 MB, comfortably within modern GPU capacity and moderately higher than the baseline's 500 MB. Processing a batch of 16 subjects is feasible, taking about 3.2 seconds and 4.8 GB of memory. Crucially, the model's parameter count is comparable to standard 3D CNNs, confirming that the added graph complexity does not cause a prohibitive increase in size. These metrics confirm the framework's viability for research and potential clinical application.

TABLE III. MODEL COMPLEXITY AND INFERENCE SPEED FOR THE PROPOSED METHOD VS. BASELINES.

Model	# Parameters (M)	GFLOPs	Memory (MB)	Inference Time (ms)
Proposed LGNN	15.2	102	800	200
CNN–LSTM (image + time)	12.7	89	500	150
LSTM (ROI features only)	0.5	0.5	100	20

In terms of computational cost, the proposed LGNN requires slightly more resources than a conventional CNN-LSTM, due primarily to the graph convolution and attention operations on longitudinal data. Overhead is modest relative to CNN-LSTM and justified by performance gains. The model size (15 million parameters, ~60 MB on disk if stored in single precision) is well within the norms for modern deep networks and can be reduced further via compression techniques if needed. We also measured training time: roughly 2.5 hours per epoch on the training set, converging in ~50 epochs (this is manageable with current hardware but could be accelerated with multiple GPUs or mixed-precision training). At test time, the method's ~5 subjects/second throughput suggests it could scale to large cohort evaluations. In summary, the LGNN offers a favorable balance between accuracy and efficiency, achieving superior results on AD progression prediction with a computational footprint that remains practical for real-world applications.

V. CONCLUSION

In conclusion, we presented a novel spatio-temporal graph neural network architecture that integrates CNN-based patch-level feature encoding with attention-driven graph modeling to capture longitudinal neuroimaging patterns of Alzheimer's disease progression. By modeling localized atrophy and inter-regional relationships over time, this approach effectively leverages longitudinal neuroimaging data. The proposed method achieved high prognostic performance in forecasting MCI-to-AD conversion, attaining an AUC of 0.89 and a concordance index of 0.84, along with high sensitivity. Interpretability analysis revealed that the model's attention weights highlight regions such as the hippocampus and posterior cingulate cortex, providing biological plausibility consistent with known Alzheimer's pathology. Furthermore, the integrated patch-level and graph-based design demonstrated robust generalizability across diverse imaging cohorts, suggesting the model's utility in varied clinical settings. Collectively, these findings underscore

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the translational potential of our approach for individualized risk stratification, early therapeutic trial enrollment, and integration into clinical diagnostic workflows.

VI. Discussion

This study was developed and evaluated using the ADNI cohort, which is well-curated and widely used but does not fully reflect the heterogeneity of real-world clinical settings (e.g., scanner protocols, recruitment bias, demographic variation). As a result, the reported performance should be interpreted as evidence of efficacy within the ADNI dataset, rather than as definitive proof of generalizability.

Additionally, although the proposed framework leverages longitudinal observations, it does not explicitly account for irregular time intervals between visits. Temporal dependencies are modeled through visit ordering and inter-visit transitions, which may not fully capture the variability in follow-up schedules.

Lastly, some sections of the manuscript could be refined further for clarity and conciseness to improve overall readability.

To strengthen external validity, we plan to evaluate the proposed model on independent cohorts, such as multi-site datasets with varying acquisition protocols, and to conduct site-held-out evaluations to quantify robustness to domain shift. We also intend to assess model calibration and examine clinically relevant operating points, such as high-sensitivity screening thresholds, to better characterize practical utility beyond standard discrimination metrics.

The proposed temporal graph connects consecutive visits and models disease progression as a sequence of transitions. While this approach captures longitudinal trends, it does not explicitly encode the time intervals between visits, which can vary significantly across subjects. In future work, we plan to incorporate Δt as an edge feature or through positional or time encoding, and evaluate whether explicitly modeling time intervals improves performance under sparse or irregular follow-up. This will be tested through stratified experiments comparing dense and sparse visit histories, as well as through controlled ablation studies.

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